Robust hedging of forward starting options

Let S_0 denote the initial stock price and let $X = S_{T_1}$ and $Y = S_{T_2}$ for $0 < T_1 < T_2$. We wish to maximize or minimize the price of a **forward-starting** option with payoff $|Y - X|$, over *all* **martingale** models $(S_t)_{t \geq 0}$ (not necessarily continuous), subject to

$$X \sim \mu \quad \text{and} \quad Y \sim \nu \tag{1}$$

i.e. that S_{T_1} and S_{T_2} have given distributions μ and ν respectively (in practice these distributions can be inferred from observed **call option** prices since if S is a random variable with density μ then $\frac{\partial^2}{\partial K^2}\mathbb{E}((S-K)^+) = \mu(K)$, see section 3.1 of the FM14 notes on the Breeden-Litzenberger formula for a simple proof of this). Note we are not choosing a specific model like Heston or SABR here.

The martingale condition implies in particular that $\mathbb{E}(X) = \mathbb{E}(Y) = S_0$ so

$$\int_{-\infty}^{\infty} x\mu(dx) \quad = \quad \int_{-\infty}^{\infty} y\nu(dy) \quad = \quad S_0$$

and without loss of generality we take $S_0 = 0$, since we can always add or subtract a constant to the process S. A sufficient condition for the existence of a martingale satisfying the marginal constraints in (1) is the **convex ordering** condition:

$$\mathbb{E}((X-K)^+) \quad \leq \quad \mathbb{E}((Y-K)^+)$$

for all $K \in \mathbb{R}$, which essentially says that for fixed strike K, a European call option with the larger maturity T_2 is worth at least as much as a call option with the smaller maturity T_1. To see why this is a necessary condition, note that

$$\begin{aligned}
\mathbb{E}((S_{T_2}-K)^+) \quad &= \quad \mathbb{E}(\mathbb{E}((S_{T_2}-K)^+|S_{T_1})) \qquad \text{(from the \textbf{tower property} in FM01)} \\
&\geq \quad \mathbb{E}(\mathbb{E}(S_{T_2}-K|S_{T_1})^+) \\
&\qquad \text{(from the \textbf{conditional Jensen inequality} applied to the convex function } f(S) = (S-K)^+) \\
&= \quad \mathbb{E}((S_{T_1}-K)^+)\,.
\end{aligned}$$

It is much harder to show that this is a sufficient condition; this follows from a deeper result called **Strassen's theorem**.

We can re-write the martingale condition as

$$\mathbb{E}((Y-X)1_{X\in A}) \quad = \quad 0$$

for all $A \in \mathcal{B}(\mathbb{R})$ (the collection of Borel sets on $\mathbb{R}$, see FM01). To see why this condition is necessary, using the tower property of conditional expectations from FM01 we note that

$$\mathbb{E}((Y-X)1_{X\in A}) \quad = \quad \mathbb{E}(\mathbb{E}((Y-X)1_{X\in A}|X)) \quad = \quad \mathbb{E}((\mathbb{E}(Y|X)-X)1_{X\in A}) \quad = \quad 0$$

using the martingale condition $\mathbb{E}(Y|X) = X$. The sufficiency of this condition follows from the rigorous definition of conditional expectation (see e.g. FM01).

We now let $\mathcal{M}(\mu,\nu)$ denote the collection of **martingale joint distributions** ρ for X and Y such that $X \sim \mu$, $Y \sim \nu$ and $\mathbb{E}^\rho(Y|X) = X$. Then we can write the **maximization** problem as

$$P \quad := \quad \max_{\rho\in\mathcal{M}(\mu,\nu)}\mathbb{E}^\rho(|X-Y|) \quad = \quad \max_{\rho\in\mathcal{M}(\mu,\nu)}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}|y-x|\rho(dx,dy) \tag{2}$$

where $\mathbb{E}^\rho(.)$ means compute the expectation under the assumption that X and Y have joint distribution ρ. This is known as the **primal** problem and as the **Hobson-Neuberger** problem. We can also consider the associated **dual** problem:

$$D \quad := \quad \min_{\alpha,\beta,\gamma\,:\,|x-y|\leq\alpha(x)+\beta(y)+\gamma(x)(y-x)\,\forall x,y\in\mathbb{R}}\Big(\int_{-\infty}^{\infty}\alpha(x)\mu(dx) \;+\; \int_{-\infty}^{\infty}\beta(y)\nu(dy)\Big)$$

where we are minimizing the cost of the superhedge for the three functions $\alpha(x)$, $\beta(y)$ and $\gamma(x)$. In words the dual problem here is: compute the cheapest **superhedge** portfolio (α,β,γ) i.e. such that

$$|x-y| \quad \leq \quad \alpha(x)+\beta(y)+\gamma(x)(y-x) \tag{3}$$

is satisfied for all $x, y \in \mathbb{R}$, i.e. the superhedge works in all scenarios. This means we are allowed to buy an arbitrary European option payoff α on S_{T_1}, an arbitrary European option payoff β on S_{T_2} and buy/sell an arbitrary amount of stock $\gamma(S_{T_1})$ at time T_1. In many cases, $P = D$ which can be proved using **duality theory**. To see why $P \leq D$, first consider any superreplicating strategy (α, β, γ) and consider any admissible model ρ in $\mathcal{M}(\mu, \nu)$ (we know such a model exists from the convex ordering condition). Then

$$\begin{aligned}
\mathbb{E}^\rho(|X - Y|) &\leq \mathbb{E}^\rho(\alpha(X)) + \mathbb{E}^\rho(\beta(Y)) + \mathbb{E}^\rho(\gamma(X)(Y - X)) \\
&= \mathbb{E}^{X\sim\mu}(\alpha(X)) + \mathbb{E}^{Y\sim\nu}(\beta(Y)) + 0 \\
&= \int_{-\infty}^{\infty} \alpha(x)\mu(dx) + \int_{-\infty}^{\infty} \beta(y)\nu(dy) \qquad (4)
\end{aligned}$$

where we have used that $X \sim \mu$, $Y \sim \nu$ and the martingale condition in the final line.

If we now let $\mathcal{A}$ denote the space of all admissible superreplicating portfolios of the form in (10). Then taking the max over all ρ on the left hand side of (4) and then the min over all portfolios in $\mathcal{A}$ on the right hand side we obtain the so-called weak duality

$$\max_{\rho\in\mathcal{M}(\mu,\nu)}\mathbb{E}^\rho(|X - Y|) = P \leq \min_{(\alpha,\beta,\gamma)\in\mathcal{A}} \int_{-\infty}^{\infty} \alpha(x)\mu(dx) + \int_{-\infty}^{\infty} \beta(y)\nu(dy) = D\,.$$

We can write the corresponding **minimization** problem as

$$P := \min_{\rho\in\mathcal{M}(\mu,\nu)}\mathbb{E}^\rho(|X - Y|) = \min_{\rho\in\mathcal{M}(\mu,\nu)} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |y - x|\rho(dx, dy) \qquad (5)$$

(this is the Hobson-Neuberger problem) for which the associated **dual** problem is now:

$$D := \max_{\alpha,\beta,\gamma\,:\,|x-y|\geq\alpha(x)+\beta(y)+\gamma(x)(y-x)\forall x,y}\Big(\int_{-\infty}^{\infty} \alpha(x)\mu(dx) + \int_{-\infty}^{\infty} \beta(y)\nu(dy)\Big)\,.$$

and for this problem we can easily verify that the weak duality is $P \geq D$. Note this is now computing the most expensive **subhedge** for $|x - y|$, or equivalently the cheapest superhedge for $-|y - x|$.

The explicit Hobson-Klimmek solution

Assume μ and ν both have densities, and $\mu(x) \geq \nu(x)$ for $x \in E = [a, b]$ and $\mu(x) \leq \nu(x)$ otherwise, and assume $\nu(x) > 0$ if $x \in [\ell, r]$, and zero otherwise, with $[a, b] \subset [\ell, r]$. For the minimization problem in (5), the minimizing "model" can then be characterized as follows:

- Let $X \sim \mu$ (which we can simulate by setting $X = F_X^{-1}(U)$ where $U \sim U[0, 1]$ if X has a density, see first FM02 lecture). Then set $Y = X$ with probability $\frac{\nu(X)}{\mu(X)}$ if $X \in E$, or probability 1 if $X \notin E$.
- If $X \in E$ and $Y \neq X$, then there are only two possibilities: $Y = q(X) > X$ or $Y = p(X) < X$ for two decreasing functions q and p such that $q : E \to [b, r]$ and $p : E \to [\ell, a]$ (see first graph below). $\mathbb{P}(Y = q(x)|X = x, Y \neq X) = \mathfrak{q}(x)$, where $\mathfrak{q}(x)$ is uniquely determined by the martingale condition:

  $$\mathbb{E}(Y|X = x) = x = \mathfrak{q}(x)q(x) + (1 - \mathfrak{q}(x))p(x)\,.$$

  We can use a **Bernoulli** random variable Z_1 with success rate $\frac{\nu(X)}{\mu(X)}$ to decide whether $X = Y$ (or not), and if $X \neq Y$ we can use an additional independent Bernoulli random variable Z_2 with success rate $\mathfrak{q}(X)$ to determine whether $Y = q(X)$ or $p(X)$. Z_1 and Z_2 can in turn be simulated using a $U[0, 1]$ random variable (why?)
- Thus we see the optimal model is a **trinomial tree** from time T_1 to T_2.
- q and p can be computed as the explicit solution to the pair of **coupled ODEs**:

  $$q'(x) = \frac{(p(x) - x)(\mu(x) - \nu(x))}{(q(x) - p(x))(\nu(q(x)) - \mu(q(x)))} \quad , \quad p'(x) = \frac{(x - q(x))(\mu(x) - \nu(x))}{(q(x) - p(x))(\nu(p(x)) - \mu(p(x)))}$$

  for $x \in E$ with $p(a) = a$ and $q(a) = r$ (see first plot below), which can be solved numerically using an Euler scheme (note one can incur numerical problems if $\nu(q(x)) - \mu(q(x)) = 0$ or $\nu(p(x)) - \mu(p(x)) = 0$, which may require adjustments to the code).
- To verify that $Y \sim \nu$, note that for $y > b$

$$\begin{aligned}
\mathbb{P}(Y \geq y) &= 1 - F_\mu(y) + \int_a^{q^{-1}(y)} \frac{x - p(x)}{q(x) - p(x)}(\mu(x) - \nu(x))dx \\
&= 1 - F_\mu(y) + \int_a^{q^{-1}(y)} q'(x)(\mu(q(x)) - \nu(q(x)))dx \\
&= 1 - F_\mu(y) + (F_\mu(q(x)) - F_\nu(q(x)))|_a^{q^{-1}(y)} \\
&= 1 - F_\mu(y) + F_\mu(y) - F_\nu(y) - (F_\mu(r) - F_\nu(r)) = 1 - F_\nu(y)\,. \qquad (6)
\end{aligned}$$

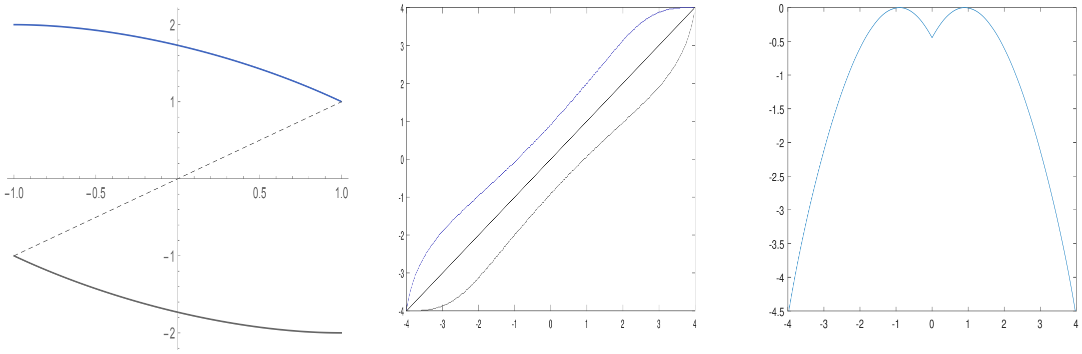

Figure 1: On the left we have plotted $q(x)$ (blue) and $p(x)$ (solid grey) for the case when $X \sim U[-1,1]$ and $Y \sim U[-2,2]$ for the minimization problem in (5), where $q(x) = \frac{1}{2}(\sqrt{12-3x^2}-x)$, $p(x) = \frac{1}{2}(-\sqrt{12-3x^2}-x)$, and the dashed line is $y = x$, and we find that $P = D = 0.5931$. In the middle we have plotted $q(x)$ (blue), $p(x)$ (grey) and $y = x$ (black) for the maximization problem in (2) for the case when $X \sim N(0,1)$ and $Y \sim N(0,2)$ i.e. the marginals of Brownian motion at times 1 and 2, which we know are in convex order since W is a martingale, and we find (using Matlab) that $P = D \approx .958$. On the right we have plotted $L(x,y) = |x-y| - (\alpha(x) + \beta(y) + \gamma(x)(y-x))$ for the same problem as the middle graph, i.e. the forward-starter terminal payoff minus the superhedge payoff which is non-positive since we have a superhedge here, and note that L is tangential to zero at $p(x) = -.92$ and $q(x) = .92$ (we have set $x = 0$ for this example, but we can consider any x-value).

where F_μ and F_ν are the distribution functions of μ and ν respectively. We can perform a similar computation for $y < a$ and $y \in [a,b]$.

- We can compute the minimal price in terms of p, q, μ and ν as

$$\begin{aligned}\mathbb{E}(|Y-X|) &= \int_a^b ((q(x)-x)\frac{x-p(x)}{q(x)-p(x)} + (x-p(x))\frac{q(x)-x}{q(x)-p(x)})(1-\frac{\nu(x)}{\mu(x)})\mu(x)dx \\ &= 2\int_a^b \frac{(x-p(x))(q(x)-x)}{q(x)-p(x)}(\mu(x)-\nu(x))dx\,. \end{aligned} \tag{7}$$

- For the example in Figure 1, these simplify to $q'(x) = \frac{p(x)-x}{q(x)-p(x)}$, $p'(x) = \frac{x-q(x)}{q(x)-p(x)}$ for $x \in [-1,1]$, which can be solved explicitly.

The Hobson-Neuberger solution

For the maximization problem in (2), the optimal solution is such that $Y = q(X)$ or $Y = p(X)$ where q and p are both **increasing functions** with $q(x) > x$ and $p(x) < x$ (see second plot below). This is a **binomial** tree model from T_1 to T_2. For the special case when $X \sim U[-1,1]$ and $Y \sim U[-2,2]$, $q(x) = x+1$ and $p(x) = x-1$. See second plot below for a more interesting numerical example for this problem.

The discrete case

If we assume X and Y have discrete distributions with $\mathbb{P}(X = x_i) = \mu_i$ for $i = 1,\dots,m$ and $\mathbb{P}(Y = y_j) = \nu_j$ for $j = 1,\dots,n$ (or we approximate the true μ and ν in this way), then we can re-write the problem as a finite-dimensional **linear programming** problem:

$$P \quad := \quad \max_{\rho_{ij}} \sum_{i=1}^m \sum_{j=1}^n \rho_{ij}|x_i - y_j| \tag{8}$$

subject to the constraints:

$$\begin{aligned}\sum_{j=1}^n \rho_{ij} &= \mu_i \quad i = 1,\dots,m, \quad \sum_{i=1}^m \rho_{ij} = \nu_j \quad j = 1,\dots,n, \\ \rho_{ij} &\geq 0 \quad i = 1,\dots,m, \quad j = 1,\dots,n, \quad \sum_{j=1}^n \rho_{ij}(y_j - x_i) \;=\; 0 \quad i = 1,\dots,m. \end{aligned} \tag{9}$$

where $\rho_{ij} = \mathbb{P}(X = x_i, Y = y_j)$. The associated dual problem is

$$\min_{\alpha,\beta,\gamma}\left(\sum_{i=1}^{m} \alpha_i \mu_i \,+\, \sum_{j=1}^{n} \beta_j \nu_j\right)$$

where $\alpha \in \mathbb{R}^m$, $\beta \in \mathbb{R}^n$ and $\gamma \in \mathbb{R}^m$ are vectors to be solved for, subject to the superhedging constraint:

$$|x_i - y_j| \quad \leq \quad \alpha_i \,+\, \beta_j \,+\, \gamma_i(y_j - x_i)$$

for all $i = 1, \ldots, m$, $j = 1, \ldots, n$. Matlab can often solve these types of problems very quickly numerically, using the **linprog** command. To be consistent with `linprog`, we first have to re-write the inequality in the form

$$-\alpha_i \,-\, \beta_j \,-\, \gamma_i(y_j - x_i) \quad \leq \quad -|x_i - y_j|\,.$$

We can then re-write this constraint in "stacked" matrix form $\mathbf{Ax} \leq \mathbf{b}$ as

$$-\begin{bmatrix} 1 & 0 & .... & 1 & 0 & .... & 0 & y_1 - x_1 & ... & 0 \\ 1 & 0 & ... & 0 & 1 & ... & 0 & y_2 - x_1 & ... & 0 \\ ... & & & & & & & & & \\ 1 & 0 & ... & 0 & 0 & ... & 1 & y_n - x_1 & ... & 0 \\ 0 & 1 & .... & 1 & 0 & .... & 0 & y_1 - x_2 & ... & \\ 0 & 1 & 0 & 0 & 1 & ... & 0 & y_2 - x_2 & ... & \\ ... & & & & & & & & & \\ 0 & 1 & 0 & 0 & 0 & ... & 1 & y_n - x_2 & ... & \\ ... & & & & & & & & & \\ 0 & ... & 1 & 0 & 0 & ... & 1 & 0 & ... & y_n - x_m \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ ... \\ \alpha_m \\ \beta_1 \\ \beta_2 \\ ... \\ \beta_n \\ \gamma_1 \\ \gamma_2 \\ ... \\ \gamma_m \end{bmatrix} \quad \leq \quad - \begin{bmatrix} |x_1 - y_1| \\ |x_1 - y_2| \\ ... \\ |x_1 - y_n| \\ |x_2 - y_1| \\ |x_2 - y_2| \\ ... \\ |x_2 - y_n| \\ ... \\ |x_m - y_n| \end{bmatrix}$$

where we have stacked α, β, γ together as a single vector $\mathbf{x}$ so as to be consistent with the way the Matlab `linprog` function expects its input arguments. If e.g. $n = m = 2$, this matrix constraint simplifies to:

$$-\begin{bmatrix} 1 & 0 & 1 & 0 & y_1 - x_1 & 0 \\ 1 & 0 & 0 & 1 & y_2 - x_1 & 0 \\ 0 & 1 & 1 & 0 & 0 & y_1 - x_2 \\ 0 & 1 & 0 & 1 & 0 & y_2 - x_2 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \\ \gamma_1 \\ \gamma_2 \end{bmatrix} \quad \leq \quad - \begin{bmatrix} |x_1 - y_1| \\ |x_1 - y_2| \\ |x_2 - y_1| \\ |x_2 - y_2| \end{bmatrix}$$

If we additionally include a tradeable option with payoff $|X - Y|$ and price P_1 in order to superhedge the non-at-the-money payoff $|kY - X|$, then for $n = m = 2$ the dual objective becomes

$$\min_{\alpha,\beta,\gamma,\delta}\left(\sum_{i=1}^{2} \alpha_i \mu_i \,+\, \sum_{j=1}^{2} \beta_j \nu_j \,+\, \delta P_1\right)$$

and the matrix constraint becomes

$$-\begin{bmatrix} 1 & 0 & 1 & 0 & y_1 - x_1 & 0 & |x_1 - y_1| \\ 1 & 0 & 0 & 1 & y_2 - x_1 & 0 & |x_1 - y_2| \\ 0 & 1 & 1 & 0 & 0 & y_1 - x_2 & |x_2 - y_1| \\ 0 & 1 & 0 & 1 & 0 & y_2 - x_2 & |x_2 - y_2| \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \\ \gamma_1 \\ \gamma_2 \\ \delta \end{bmatrix} \quad \leq \quad - \begin{bmatrix} |x_1 - ky_1| \\ |x_1 - ky_2| \\ |x_2 - ky_1| \\ |x_2 - ky_2| \end{bmatrix}$$

We can similarly re-write the equality constraints for the primal problem in (9) as:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ y_1 - x_1 & y_2 - x_1 & 0 & 0 \\ 0 & 0 & y_1 - x_2 & y_2 - x_2 \end{bmatrix} \begin{bmatrix} \rho_{11} \\ \rho_{12} \\ \rho_{21} \\ \rho_{22} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \nu_1 \\ \nu_2 \\ 0 \\ 0 \end{bmatrix}$$

and the inequality constraint is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \rho_{11} \\ \rho_{12} \\ \rho_{21} \\ \rho_{22} \end{bmatrix} \quad \geq \quad \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

If we wish to add an additional tradeable option with payoff $|X-Y|$ and price P_1, we add an additional row of the form:

$$\begin{bmatrix}|x_1-y_1| & |x_1-y_2| & |x_2-y_1| & |x_2-y_2|\end{bmatrix}\begin{bmatrix}\rho_{11}\\ \rho_{12}\\ \rho_{21}\\ \rho_{22}\end{bmatrix} \quad = \quad \begin{bmatrix}P_1\end{bmatrix}$$

P_1 has to lie in $[\underline{P}, \bar{P}]$, where $\underline{P}$ and $\bar{P}$ are the primal values for (5) and (2) respectively, or else no model exists for which $\mathbb{E}^\rho(|X-Y|) = P_1$ and $X \sim \mu$ and $Y \sim \nu$. If we instead have a bid-offer spread ε for this contract (i.e. ask price is $P_1+\varepsilon$ and bid price is $P_1-\varepsilon$), then this row is replaced with

$$\begin{bmatrix}|x_1-y_1| & |x_1-y_2| & |x_2-y_1| & |x_2-y_2|\end{bmatrix}\begin{bmatrix}\rho_{11}\\ \rho_{12}\\ \rho_{21}\\ \rho_{22}\end{bmatrix} \quad \leq \quad \begin{bmatrix}P_1+\varepsilon\end{bmatrix}$$

and

$$\begin{bmatrix}|x_1-y_1| & |x_1-y_2| & |x_2-y_1| & |x_2-y_2|\end{bmatrix}\begin{bmatrix}\rho_{11}\\ \rho_{12}\\ \rho_{21}\\ \rho_{22}\end{bmatrix} \quad \geq \quad \begin{bmatrix}P_1-\varepsilon\end{bmatrix}$$

which can also be added as an inequality constraint in Matlab. We can similarly adjust the dual problem to account for ε.

Things to investigate:

- Consider the **non-at-the-money** case where the payoff is $|ky-x|$ for $k \neq 1$. We can add the $k=1$ contract as a tradeable instrument with a given price which lies between the lower and upper bounds discussed above, and then compute the max/min price and corresponding subhedge/superhedge for an additional forward starter with $k \neq 1$. In this case the superhedge looks like

  $$|ky-x| \quad \leq \quad \alpha(x)+\beta(y)+\gamma(x)(y-x)\ +\ \delta|y-x|$$

  where $\delta \in \mathbb{R}$ is the amount of the $k=1$ contract that we buy at time zero.

- Work with only finite tradeable calls and puts as is the case in practice, and possibly add bid/offer spreads, i.e. **transaction costs** to their prices and/or the trade in S itself at time T_1.

- **Barrier options**: μ is the joint law of X_T and its maximum $\bar{X}_T$ at time T only if $X_T \leq \bar{X}_T$ almost surely and the Rogers condition $\mathbb{E}((X_T-a)1_{\bar{X}_T \geq a}) \quad = \quad 0$ is satisfied for all relevant a. This constraint could be added into the Matlab constraints with $\bar{X}$ replacing Y, and we can dispense with the second maturity T_2. One can then find e.g. the smallest or largest price of a barrier option paying $(X_T-K)^+1_{\bar{X}_T>B}$ subject to a finite/infinite set of put/call options on X_T and options on $\bar{X}_T$, e.g. **One-Touch options**.

- **VIX options**